

Security Validation and Compliance Engineering Assignment Solution

Student Name: Hassan Raza

Course: Security Validation and Compliance Engineering

Section A – Validation Criteria and Checklist Development

Q1. Validation Criteria Engineering

1. Measurable Validation Criteria

System Feature	Stakeholder Expectation	Validation Criteria
Student Login	Secure and fast login	Login success within 3 seconds with valid credentials
Assignment Submission	Reliable file upload	Files uploaded successfully with confirmation message
Attendance Tracking	Accurate attendance records	Attendance data updated without duplication
Faculty Dashboard	Easy monitoring and reporting	Dashboard loads correctly with updated student data
Activity Logging	System activity tracking	Logs stored with timestamp and user details

2. Structured Validation Checklist

Checklist ID	Feature	Validation Test	Expected Result	Status
VC-01	Student Login	Enter valid username/password	User logged in successfully	Pass/Fail
VC-02	Assignment Submission	Upload assignment file	File uploaded with success message	Pass/Fail
VC-03	Attendance Tracking	Mark attendance	Attendance saved correctly	Pass/Fail
VC-04	Faculty Dashboard	Open dashboard	Dashboard displays student records	Pass/Fail
VC-05	Activity Logging	Perform system activity	Activity recorded in logs	Pass/Fail

Section B – Continuous Validation Automation using CI Pipelines

Q2. Validation Automation using GitHub Actions

1. Role of CI Pipelines in Continuous Validation

Continuous Integration (CI) pipelines automatically validate software whenever new code is committed. They help teams detect bugs early, execute automated tests, generate validation reports, and maintain software quality. CI pipelines also provide audit evidence for compliance and security validation. The purpose is to install CI pipeli

2. GitHub Actions Workflow

```
name: Validation Pipeline

on:
  push:
    branches:
      - main

jobs:
  validation:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4

      - name: Setup Node.js
        uses: actions/setup-node@v4
        with:
          node-version: '18'

      - name: Install Dependencies
        run: npm install

      - name: Execute Tests
        run: npm test

      - name: Generate Validation Logs
        run: |
          mkdir validation-logs
          echo "Validation completed successfully" > validation-logs/report.txt

      - name: Upload Validation Artifacts
        uses: actions/upload-artifact@v4
        with:
          name: validation-evidence
          path: validation-logs/
```

Section C – Security Validation and Vulnerability Assessment

Q3. Structural Vulnerability Examination using OWASP ZAP

1. Importance of Security-Oriented Validation

Security validation ensures that software systems are protected against cyber threats and vulnerabilities. It helps maintain confidentiality, integrity, and availability of data before deployment.

2. Common Vulnerabilities

Vulnerability	Explanation
SQL Injection	Attackers inject malicious SQL queries to access or modify database data.
Cross-Site Scripting (XSS)	Malicious scripts are injected into web pages and executed in user browsers.
Broken Authentication	Weak authentication allows attackers to access user accounts illegally.

3. OWASP ZAP Vulnerability Assessment Workflow

- Configure target URL in OWASP ZAP.
- Run passive scanning to detect security issues without attacking the application.
- Perform active scanning to test vulnerabilities such as SQL Injection and XSS.
- Review risk classification (Low, Medium, High).
- Generate scan reports and validation evidence in HTML/PDF format.

Expected Deliverables

- ZAP Scan Report
- Vulnerability Severity Table
- Evidence Screenshots and Logs
- Mitigation Recommendations

Section D – Validation Risk and Compliance Documentation

Q4. Validation Risk Management and Compliance Evidence

1. Validation Risk Register

Risk	Impact	Likelihood	Mitigation
Unauthorized Access	Data breach	High	Use multi-factor authentication
Missing Audit Logs	Compliance failure	Medium	Enable automated logging
Weak Password Policy	Account compromise	High	Apply strong password rules

2. Compliance Evidence Repository Structure

/compliance-evidence

```
/validation-checklists  
/zap-scan-reports  
/github-actions-logs  
/risk-register  
/screenshots  
README.md
```